

Chapter 1

A. Indices

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$a^{-n} = \frac{1}{a^n}$$

$$a^0 = 1$$

$$(a^m)^n = a^{mn}$$

$$a^m \times b^n = a^m b^n$$

COMMON MISTAKES!

$$a^m \times a^n = a^m + a^n$$

$$a^m \div a^n = a^m - a^n$$

$$a^m \times b^m = (a+b)^m$$

$$a^m \times b^n = a^m + b^n$$

B. Surds

$$\sqrt{a} \times \sqrt{a} = a$$

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$(\sqrt{a})^m = a^{\frac{m}{2}}$$

COMMON MISTAKES!

$$\sqrt{a} \times \sqrt{b} = ab$$

$$\sqrt{a} \times \sqrt{a} = a^2$$

Conjugate/ Rationalise surds

$$\sqrt{a} \Rightarrow \sqrt{a}$$

$$b \pm \sqrt{a} \Rightarrow b \mp \sqrt{a}$$

$$\sqrt{a} + \sqrt{b} \Rightarrow \sqrt{a} - \sqrt{b}$$

$$\sqrt{b} - \sqrt{a}$$

COMMON MISTAKES!

$$\sqrt{a} + \sqrt{b} \Rightarrow -\sqrt{a} - \sqrt{b}$$

$$\frac{1}{\sqrt{a} + \sqrt{b}} = \frac{1}{\sqrt{a} + \sqrt{b}} \times \frac{\sqrt{a} - \sqrt{b}}{1}$$

c. Logarithms

$$\log a + \log b = \log(ab) \neq (\log a)(\log b)$$

$$\log a \div \log b = \log\left(\frac{a}{b}\right) \neq \frac{\log a}{\log b}$$

$$\log a^m = m \log a \neq (\log a)^m$$

$$a^{\log_a x} = x$$

$$\log_a b = \frac{\log_b b}{\log_b a} = \frac{1}{\log_b a} \neq \frac{\log_b a}{\log_b b}$$

COMMON MISTAKES!

$$\log a + \log b = \log(ab) = (\log a)(\log b)$$

$$\log a \div \log b = \log\left(\frac{a}{b}\right) = \frac{\log a}{\log b}$$

$$\log a^m = m \log a = (\log a)^m$$

$$\log_a b = \frac{\log_b a}{\log_b b}$$

D. Polar form

$$z = a + bi, \quad \bar{z} = a - bi$$

$$\text{Modulus, } r = \sqrt{a^2 + b^2}$$

Argument, θ

$$\theta = \pi - \tan^{-1}\left(\frac{b}{a}\right)$$

$$\theta = -\left(\pi - \tan^{-1}\left(\frac{b}{a}\right)\right)$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$

$$\theta = -\tan^{-1}\left(\frac{b}{a}\right)$$

POLAR FORM

$$z = r(\cos \theta + i \sin \theta)$$

in RADIAN!!

CHAPTER 2

A. Indices

- same base
- let $(3^x = a)$ to simplify the equation to be quadratic function
- factorise the function
- replace back $3^x = a$ to find values of x

B. Surds

- collects the surds on one side and numbers on other side
- Square Both Sides (SBS)

i. one surd $\sqrt{a+x} + b = 0 \Rightarrow (\sqrt{a+x})^2 = (-b)^2$

ii. two surds $\sqrt{a+x} + b = \sqrt{c+x} \Rightarrow (\sqrt{a+x} - \sqrt{c+x})^2 = (-b)^2$

iii. three surds $\sqrt{a+x} + \sqrt{b+x} = \sqrt{c+x} \Rightarrow (\sqrt{a+x} + \sqrt{b+x})^2 = (\sqrt{c+x})^2$

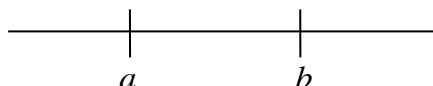
C. Logarithms

- same base
- let $(3^x = a)$ to simplify the equation to be quadratic function
- factorise the function
- replace back $3^x = a$ to find values of x
- **RECHECK!**

RECHECK!!

- Surds
- Logarithms
- Modulus
- Ln

D. Inequalities



i. Quadratic function

Graphical	Real Number Line	Table of Sign
	F actorise A ssume all factors to be positive R eal number line S ign +ve/-ve P roduct of sign A nswer	F actorise T able – Testing value - Sign - Product of sign A nswer

ii. Rational function

$$\frac{ax+b}{cx+d} < k \Rightarrow \frac{ax+b}{cx+d} - \frac{k(ax+b)}{cx+d} < 0$$

DON'T CROSS MULTIPLY!!

iii. Absolute values

a. One absolute value function

$$|ax+b| < c \Rightarrow -c < ax+b < c$$

$$ax+b < c \quad \text{and} (\cap) \quad ax+b > -c$$

$$|ax+b| > c \Rightarrow ax+b > c \quad \text{or} (\cup) \quad ax+b < -c$$

b. Two absolute value function

$$|ax+b| < |cx+d| \Rightarrow (|ax+b|)^2 < (|cx+d|)^2$$

G O L A

SQUARE BOTH SIDES!!

CHAPTER 3

ARITHMETIC	GEOMETRIC
$d = T_{n+1} - T_n$ $T_n = a + (n-1)d$ $S_n = \frac{n}{2}(2a + (n-1)d)$	$r = \frac{T_{n+1}}{T_n}$ $T_n = ar^{n-1}$ $S_n = \frac{a(1-r^n)}{1-r}, \quad r < 1$ $\frac{a(r^n-1)}{r-1}, \quad r > 1$ $S_\infty = \frac{a}{1-r}, \quad -1 < r < 1$

Binomial Expansion (BE)

a. $T_{r+1} = \binom{n}{r} (a^{n-r})(b)^r$ - independent x, x^0

- coefficient : excluding the variable

b. n is positive integer : $(a+b)^n = a^n + \binom{n}{1}a^{n-1}b^1 + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$

c. n is negative integer / fraction : $(1+b)^n = 1 + nb + \frac{n(n-1)}{2!}(b)^2 + \dots$

Simplify $(a+b)^n = a^n \left(1 + \frac{b}{a}\right)^n = a^n \left(1 + n\left(\frac{b}{a}\right) + \frac{n(n-1)}{2!}\left(\frac{b}{a}\right)^2 + \dots\right)$

Validity $\left|\frac{b}{a}\right| \leq 1$

Approximate – x given: substitute given x into LHS and RHS. Rearrange to get the approximate requested.

- x not given: use comparing method. Make sure x is in the range of validity.

CHAPTER 4

A. Basic Operation

Multiplication $A \times B$

$$\begin{array}{ccccc} A & \times & B & = & C \\ m_1 \times n_1 & & m_2 \times n_2 & & m_1 \times n_2 \end{array} \quad \boxed{n_1 = m_2}$$

NOTE: $A \times B \neq B \times A$ not commutative

B. Transpose

$$(A \pm B)^T = A^T \pm B^T$$

$$(A^T)^T = A$$

$$(AB)^T = B^T A^T$$

$$(kA)^T = kA^T$$

C. Determinant

$$|A^T| = |A|$$

$$|AB| = |A||B|$$

$$|A^2| = |A||A|$$

D. Inverse method

$$AX = B \Rightarrow X = A^{-1}B$$

(Use Adjoint method to find A^{-1})

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$|A| = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$$

$$\text{adj } A = \begin{pmatrix} +m_{11} & -m_{12} & +m_{13} \\ -m_{21} & +m_{22} & -m_{23} \\ +m_{31} & -m_{32} & +m_{33} \end{pmatrix}^T$$

PROPERTIES OF A^{-1}

$$1. (A^{-1})^{-1} = A$$

$$2. (A^T)^{-1} = (A^{-1})^T$$

$$3. (AB)^{-1} = B^{-1}A^{-1}$$

$$4. |A^{-1}| = \frac{1}{|A|}$$

$$5. AB = BA = I \Rightarrow A^{-1} = B \text{ or } B^{-1} = A$$

$$6. AB = \alpha I \Rightarrow A^{-1} = \frac{1}{\alpha} B \text{ or } B^{-1} = \frac{1}{\alpha} A$$

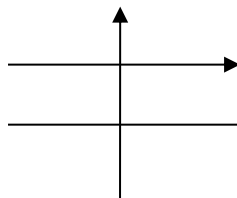
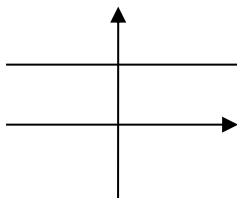
E. Gauss-Jordan method $(A|B) \Rightarrow (I|X)$

6 steps of ERO

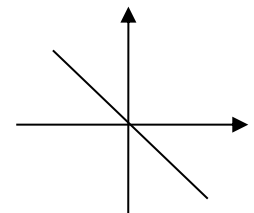
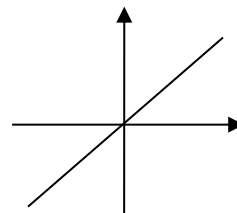
CHAPTER 5

A. Basic function

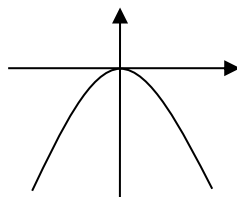
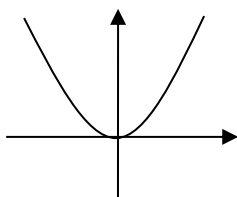
1. Constant $y = k$



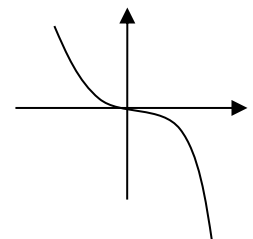
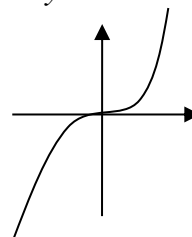
2. Linear, $y = x$



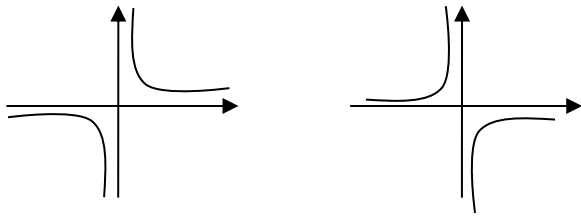
3. Quadratic $y = x^2$



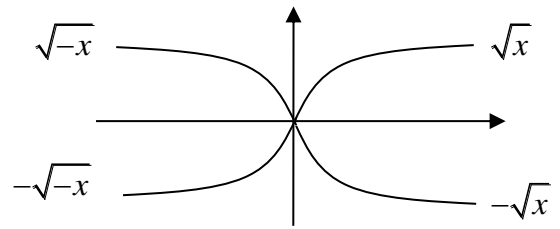
4. Cubic $y = x^3$



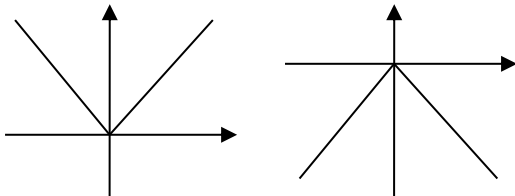
5. Reciprocal $y = \frac{1}{x}$



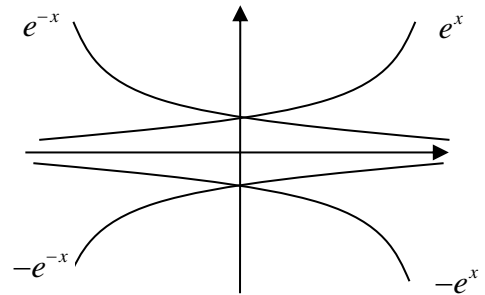
6. Surds $y = \sqrt{x}$



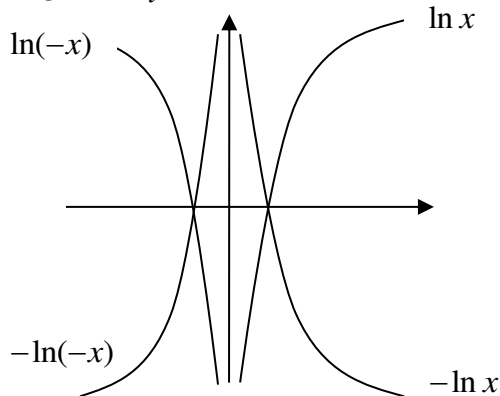
7. Absolute value $y = |x|$



8. Exponential $y = e^x$



9. Logarithm $y = \ln x$



NOTES!

$$g \circ f \neq f \circ g$$

$$g \circ f \neq g \times f$$

$$(g \circ f)^{-1} = f^{-1} g^{-1}$$

B. One-to-one function

Algebraic

$$y = \sqrt{x-2}$$

$$\text{Let } f(x_1) = f(x_2)$$

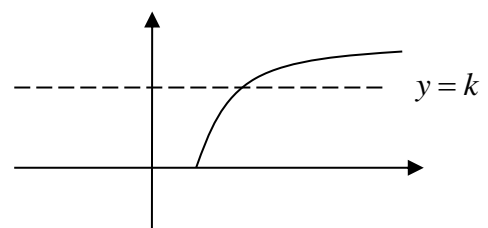
$$\sqrt{x_1-2} = \sqrt{x_2-2}$$

$$x_1-2 = x_2-2$$

$$x_1 = x_2$$

Since $x_1 = x_2$, the $f(x)$ is one-to-one function.

Graphical



Since horizontal line test cuts one point on the graph, the $f(x)$ is one-to-one function.

C. Inverse function

Algebraic

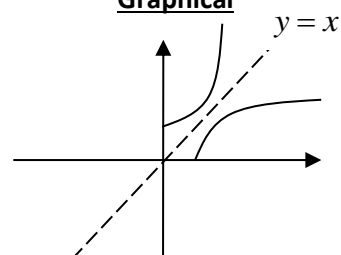
$$f(f^{-1}(x)) = x$$

$$\sqrt{f^{-1}(x)-2} = x$$

$$f^{-1}(x)-2 = x^2$$

$$f^{-1}(x) = x^2 + 2, x > 2$$

Graphical



CHAPTER 6

A. Remainder Theorem

$$D(x) = ax + b \Rightarrow R(x) = k$$

$$D(x) = ax^2 + bx + c \Rightarrow R(x) = ax + b$$

B. Factor Theorem, roots and zeros

$$x^2 - 4x + 3 = 0$$

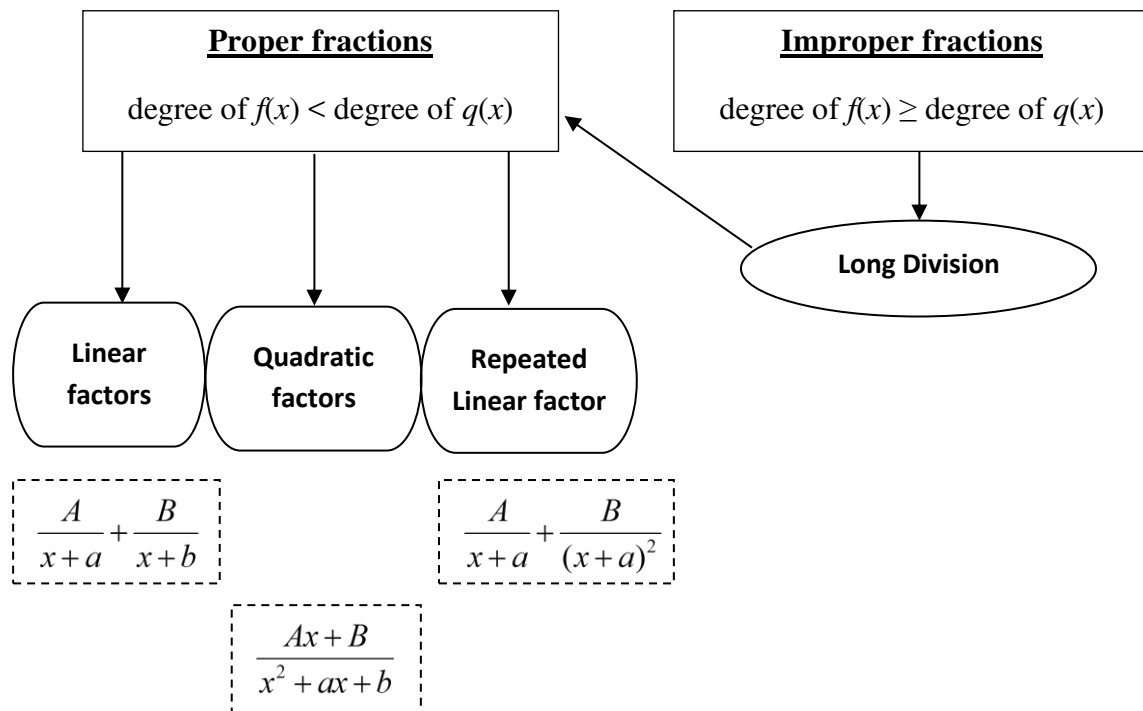
$$(x-3)(x-1) = 0 \quad \text{factor}$$

$$x = 3, x = 1 \quad \text{root}$$

$$3, 1 \quad \text{zeros}$$

Degree of n polynomial gives
 n factors, roots and zeros.

C. Partial fraction, $\frac{f(x)}{g(x)}$



CHAPTER 7

A. $t = \tan \frac{\theta}{2}$

$$\tan \theta = \tan \left(\frac{\theta}{2} + \frac{\theta}{2} \right)$$

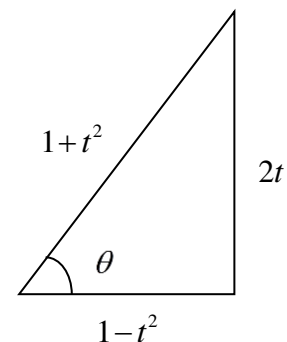
$$= \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$$

$$= \frac{2t}{1-t^2} = \frac{\text{opp}}{\text{adj}}$$

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{2t}{1+t^2}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{1-t^2}{1+t^2}$$

- substitute into the equation in term of t
- simplify and factorise t
- Need to consider the range of θ
- $0 \leq \theta \leq 2\pi \quad 0^\circ \leq \theta \leq 360^\circ$
- $0 \leq \frac{\theta}{2} \leq \pi \quad 0^\circ \leq \frac{\theta}{2} \leq 180^\circ$



B. $R\cos(\theta \pm \alpha)$, $R\sin(\theta \pm \alpha)$

$$R\cos(\theta \pm \alpha) = R\cos\theta\cos\alpha \mp R\sin\theta\sin\alpha$$

$$R\sin(\theta \pm \alpha) = R\sin\theta\cos\alpha \pm R\cos\theta\sin\alpha$$

Show all working steps!!

1. $\left. \begin{array}{l} -1 \leq \cos(\theta \pm \alpha) \leq 1 \\ -1 \leq \sin(\theta \pm \alpha) \leq 1 \end{array} \right\}$ rearrange depends on the function, $f(\theta)$
2. To find values of x , need to rearrange the range of θ .

$$0^\circ \leq \theta \leq 360^\circ$$

$$0^\circ \pm \alpha \leq \theta \pm \alpha \leq 360^\circ \pm \alpha$$
3. Consider the quadrants

C. Proving - simplify the complex/complicated part by substituting the relevant formulae

CHAPTER 8

A. Finite limits – substitute the value of x into function

$$\frac{0}{0} \text{ form}$$

Conjugate
Factorise

$$\frac{\infty}{\infty} \text{ form}$$

Divide by highest power of denominator

$$\frac{a}{\infty} = 0$$

$$\frac{a}{0} = \pm\infty$$

*** Long Division** $\frac{P(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)}$

B. Infinite limits (to find VA)

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty, \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty$$

NOTES!

$$\sqrt{x^2} = |x| = \begin{cases} x & , x \rightarrow \infty \\ -x & , x \rightarrow -\infty \end{cases}$$

C. Limits at infinity (to find HA)

$$\lim_{x \rightarrow \infty} f(x) = L, \quad \lim_{x \rightarrow -\infty} f(x) = M$$

D. Continuity

S1. $f(c)$ exists $\Rightarrow f$ is defined

S2. $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) \Rightarrow \lim_{x \rightarrow c} f(x)$ exists

S3. $\lim_{x \rightarrow c} f(x) = f(c)$

CHAPTER 9

A. 1st Principle

$$f'(x) = \frac{dy}{dx} = \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right]$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

B. Differentiation Rule

$$\frac{d}{dx}(kx^n) = knx^{n-1}$$

$$\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{vu' - uv'}{v^2}$$

$$\frac{d}{dx}(a^x) = a^x \ln a$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(\ln f(x)) = \frac{1}{f(x)} \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(u(x) \bullet v(x)) = uv' + vu'$$

$$\frac{dy}{dx} = \frac{dy}{du} \bullet \frac{du}{dx}$$

$$\frac{d}{dx}(a^{f(x)}) = a^{f(x)} \ln a \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

C. Implicit Differentiation

$$\frac{d}{dx}(y) = \frac{dy}{dx}, \quad \frac{d}{dx}(y^n) = ny^{n-1} \frac{dy}{dx}$$

Notes!

Whenever we have variable y , we will have $\frac{dy}{dx}$

STEPS

1. Differentiate each of term with respect to x
2. Rearrange and let $\frac{dy}{dx}$ as the subject

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

D. Parametric Differentiation

$$x = g(t), \quad y = f(t)$$

$$\frac{dy}{dx} = \frac{dy}{dt} \bullet \frac{dt}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

CHAPTER 10

A. Extremum points

Critical point $\longrightarrow$ Case 1: $f'(x) = 0$ (Stationary point)
 $\longrightarrow$ Case 2: $f'(x)$ is not defined

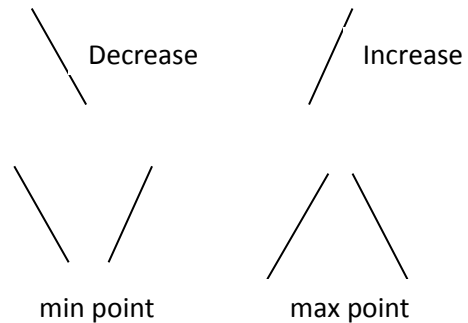
Critical number, $x = c$
 Critical value, $y = f(c)$
 Critical point, $(c, f(c))$

Method 1: 1st derivative test

S1. Find critical point

S2. To test max/min (Table)

Interval		
Test Value		
Sign of $f'(x)$		
Conclusion		



Method 2: 2nd derivative test

S1. Find critical point

S2. To test max/min - find $f''(x)$

- Substitute the values of critical number into $f''(x)$

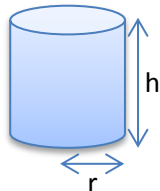
$f''(x) > 0 \longrightarrow$ Min

$f''(x) < 0 \longrightarrow$ Max

$f''(x) = 0 \longrightarrow$ Test fails. Have to do 1st derivative test

B. Formula

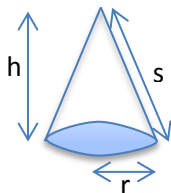
1. Cylinder



$$V = \pi r^2 h$$

$$A = 2\pi r^2 + 2\pi rh$$

2. Cone



$$V = \frac{1}{3} \pi r^2 h$$

$$A = \pi r^2 + \pi rs$$

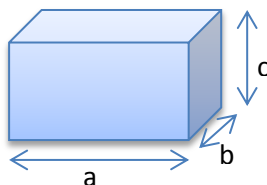
3. Sphere



$$V = \frac{4}{3} \pi r^3$$

$$A = 4\pi r^2$$

4. Rectangle



$$V = a \times b \times c$$

$$A = 2ab + 2ac + 2bc$$

CHECKLIST REVISION		✓
CHAPTER 1 : NUMBER SYSTEM		
1.	What is N, W, Z and Q for numbers?	
2.	Modulus and argument for complex number	
3.	Represent complex number in Argand diagram	
4.	Represent complex number in Polar form; $z = r(\cos \theta + i \sin \theta)$	
5.	Rules of indices	
6.	Rationalising surds	
7.	Rules of logarithms	
8.	Changing base for log	
CHAPTER 2 : EQUATIONS, INEQUALITIES AND ABSOLUTE VALUE		
1.	Solving equations involving indices	
2.	Solving equations involving logarithm	
3.	Solving equations involving surds	
4.	Solving linear inequalities	
5.	Solving quadratic inequalities	
6.	Solving rational inequalities	
7.	Solving equations with absolute value(s)	
8.	Solving inequalities with absolute value(s)	
CHAPTER 3 : SEQUENCE AND SERIES		
1.	Solving problems regarding arithmetic progression	
2.	Solving problems regarding geometric progression	
3.	Expanding binomial expansion using both formula	
4.	Approximate values of decimal using binomial expansion	
CHAPTER 4 : MATRICES AND SYSTEM OF LINEAR SYSTEM		
1.	Finding minor, cofactor, determinant and adjoint for 3X3 matrix	
2.	Finding inverse of a matrix using adjoint method	
3.	Finding inverse of a matrix using properties of inverse matrix	
4.	Finding inverse of a matrix using ERO	
5.	Solving linear system using inverse matrix	
6.	Solving linear system using Gauss Jordan method	
CHAPTER 5 : FUNCTIONS AND GRAPHS		
1.	Testing for one-to-one function (graphical and algebraic method)	
2.	Finding composite functions	
3.	Finding inverse of a function	
4.	Finding domain and range of a function(graphical and algebraic method)	
5.	Know how to sketch	
	i. Constant	
	ii. Linear	
	iii. Quadratic	
	iv. Cubic	
	v. Surds	
	vi. Modulus	
	vii. Reciprocal	
	viii. Exponential	
	ix. Logarithm	
	x. Piecewise	
CHAPTER 6 : POLYNOMIALS		
1.	Perform long division	
2.	Use remainder theorem	
3.	Use factor theorem	
4.	Finding factors, zeros and roots of a polynomial	
5.	Perform partial fraction	

CHAPTER 7 : TRIGONOMETRIC FUNCTIONS		
1.	Know the values of special angles	
2.	Know how and when to use trigonometry identities	
3.	Know how and when to use double angle, half angle, compound angle and factor formula correctly	
4.	Know how to use said formula and identities to do proving questions	
5.	Know how to solve basic trigonometry equation	
6.	Know how to solve trigonometry equation using quadratic	
7.	Know how to solve trigonometry equation using t-substitution	
8.	Know how to solve trigonometry equation using R form	
CHAPTER 8 : LIMITS AND CONTINUITY		
1.	Know how to find limit	
2.	Know how to find limit at infinity	
3.	Know how to find limit to infinity	
4.	Know the conditions for limit to exist	
5.	Finding both horizontal and vertical asymptote using limit	
6.	Know the conditions of continuity	
CHAPTER 9 : DIFFERENTIATION		
1.	Understand differentiation as a gradient (first principle)	
2.	Applying rules of differentiation	
3.	Differentiate exponential, logarithms and trigonometric functions	
4.	Performing implicit differentiation	
5.	Performing parametric differentiation	
6.	Performing higher order differentiation	
CHAPTER 10 : APPLICATION OF DIFFERENTIATION		
1.	Finding critical points	
2.	Finding turning points, stationary points, extremum points	
3.	Finding all the points above using first derivative test	
4.	Finding all the points above using second derivative test	
5.	Solving optimization problems	
6.	Solving rate of change problems	

